

GRADUATION PROJECT
DEPI (CAI4_SED2_S4)

Waste-to-Value Platform

Bridging Egypt's Informal Recycling Economy Through Digital Transformation

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Presentation Agenda

01 Introduction & Problem Statement

02 Business Plan & Financial Model

03 System Analysis & Design

04 Database & Technical Architecture

05 System Implementation

06 Testing Methodology

07 Conclusion & Future Work

01

Introduction

Problem Statement, Platform Concept & Scope

The Problem: Egypt's Broken Waste Chain

The Scale of Waste

- Egypt generates ~26 million tons of Municipal Solid Waste annually
- Formal recycling rate remains below 12%
- Corporate waste sent to landfill or sold at low prices to informal middlemen
- Recycling factories lack consistent, traceable upstream supply
- Waste collectors work informally with no market access or fair pricing
- No ESG compliance documentation for corporate clients

Our Solution

- B2B contract-based logistics platform — NOT a consumer app
- Connect corporate generators directly with industrial factories
- Algorithmic Pricing Engine: buy at 70%, sell at 110% market rate
- Formal ESG compliance reporting backed by MoE MoU
- Own 500sqm hub with 50-ton hydraulic baler
- AI-powered onboarding chatbot (Gemini)

Competitive Landscape: How We Differ

Feature	Waste-to-Value	Bekia	Go Clean	Rubicon
Target Market	Corporate B2B	Households (B2C)	Households (B2C)	Enterprises (SW only)
Pricing Model	Algorithmic (70/110)	Points/Vouchers	WhatsApp negotiated	Variable
Own Infrastructure	Yes (hub + baler)	No	No	No
ESG Compliance Docs	Yes (MoE MoU)	No	No	Partial
Factory Direct Access	Yes	No	No	No
Digital Platform	Full (React/Laravel)	App (B2C)	Phone/WhatsApp	Dashboard only

Platform Pipeline & System Users

01

Source Extraction

Corporate generators
sign contracts;
Dry-leased fleet collects

02

Hub Inbound & QA

Weigh, sort,
decontaminate;
5% threshold enforced

03

Densification & Inventory

50-ton baler;
Bale IDs assigned;
Commodity inventory

04

Outbound & Factory Offtake

24-ton FTL dispatch
to BariQ / MEPCO;
Sales invoice generated

SYSTEM USERS



Supplier
(Corporate Generator)



Recycling
Factory



Driver /
Loader



Hub
Manager



Super
Admin

System Objectives & Key Features

Algorithmic Pricing

Buy at 70%, sell at 110% — zero bidding, locked-in 40pt spread

Quality Control

5% contamination threshold enforced digitally at every hub intake

Role-Based Access

5 distinct roles: Supplier, Factory, Driver, Hub Manager, Super Admin

Live Analytics

Real-time dashboards: tonnage, fleet, inventory, margin performance

Split Logistics

Inbound: dry-leased fleet; Outbound: subcontracted 24-ton FTL flatbeds

AI Onboarding

Gemini-powered chatbot for contracts, FAQs, and waste classification

Technology Stack & Methodology

Waterfall Development Methodology

Phase 1: Requirements Gathering & Analysis

Phase 2: System Design (UML: Use Case, Activity, Sequence)

Phase 3: Implementation — React + PHP/Laravel + Gemini AI

Phase 4: Testing — Cross-browser & mobile (3G/4G)

Phase 5: Deployment — Secure Cloud (AWS)

Phase 6: Maintenance & Future Scale-Up

Technology Stack

Frontend: React.js (role-based responsive dashboards)

Backend: PHP / Laravel (REST API, RBAC, business logic)

Database: SQL Server

AI: Gemini API (chatbot — onboarding & support)

DevTools: VS Code, GitHub, Visual Paradigm (UML)

Browser: Chrome, Edge, Safari (cross-browser tested)

02

Business Plan

Market Strategy, Financial Projections & SWOT

Business Overview: Vision & Mission

VISION

To become Egypt's premier B2B infrastructure layer for the corporate waste management sector — systematically diverting commercial recyclables from landfill into a formal, traceable, algorithmically priced circular economy pipeline.

MISSION

Acquire pre-segregated recyclables from corporate generators at algorithmically determined rates, process them in our own aggregation hub, and deliver mill-ready baled commodities to tier-one industrial factories — providing formal ESG compliance under MoE MoU.

LLC

Legal Form

**Greater
Cairo**

Year 1 Area

**500
sqm**

Aggregation Hub

50-ton

Hydraulic Baler

**3
Trucks**

Dry-Leased Fleet

SWOT Analysis

STRENGTHS

- First B2B contract-based waste logistics platform in Egypt targeting corporate generators
- Algorithmic Pricing Engine: guaranteed 40pt spread (70% buy / 110% sell)
- MoE MoU provides regulatory credibility + formal ESG reporting
- Own processing infrastructure (500 sqm hub, 50-ton baler, 1-ton scale)

OPPORTUNITIES

- Law 202/2020: mandatory corporate waste tracking creates instant demand
- ESG reporting requirements from multinational tenants of malls and hotels
- PET/paper import restrictions increase domestic recycled material demand
- Hub model easily replicable across Alexandria, Mansoura, Upper Egypt

WEAKNESSES

- Heavy upfront CAPEX (558,250 EGP) before first revenue
- Dependency on factory payment discipline (Net 14 float risk)
- Small founding team (6) managing 13 operational staff
- Single hub creates geographic concentration risk

THREATS

- Factory payment delays (Net 30+) could severely impact cash flow
- Informal brokers may attempt to undercut on price
- Global commodity price volatility impacts the margin spread
- Regulatory risk: MoE MoU terms could be revised

Start-up Capital

2.6.1 Start-up Capital

Table 2.6: Start-up Capital Details (CAPEX)

Item	Specification	Amount (EGP)
Legal & Regulatory	GAFI registration, WMRA license, MoE MoU setup	50,000
Custom Tech & AI	Web platform, role-based dashboards, Gemini AI chatbot, security	50,000
Admin Hardware	6 Laptops, monitors, routers, office furniture	75,000
Heavy Hub Equipment	50-Ton Baler (350k) + 1-Ton Scale (15k) + 5 % VAT	383,250
Total CAPEX		558,250

Total Working Capital & Start-up Capital

2.6.2 Total Working Capital & Start-up Capital

Table 2.7: Total Working Capital & Start-up Capital

Category	Description	Amount (EGP)
Fixed OPEX (×12 months)	HR payroll, hub rent, cloud, utilities (211,300 /mo)	2,535,600
Variable Logistics (×12)	Truck lease, fuel, consumables, freight (122,000/mo)	1,464,000
Marketing (×12 months)	B2B sales, LinkedIn ABM, events (12,500 /mo)	150,000
Total Working Capital		4,149,600
Total Start-up Capital	CAPEX + Working Capital	4,707,850

Profit Margin

2.7.3 Profit Margin

Table 2.14: Profit Margin

Item	Q1 (Jan– Mar)	Q2 (Apr– Jun)	Q3 (Jul– Sep)	Q4 (Oct– Dec)	Total Year /
Total Revenues	3,861,000	3,861,000	3,861,000	3,861,000	15,444,000
Material Acquisition	2,457,000	2,457,000	2,457,000	2,457,000	9,828,000
Gross Margin	1,404,000	1,404,000	1,404,000	1,404,000	5,616,000
Operating Expenses	1,037,400	1,037,400	1,037,400	1,037,400	4,149,600
Profit Before Tax	366,600	366,600	366,600	366,600	1,466,400
Taxes (22.5%)					329,940
Profit After Tax					1,136,460

Balance Sheet

2.7.5 Balance Sheet

Table 2.16: Balance Sheet

Assets	Value (EGP)	Liabilities & Owner's Equity	Value (EGP)
Cash	5,286,060	Liabilities	0
Total Current Assets	5,286,060	Total Liabilities	0
Tech & AI Dashboard	50,000	Owner's Equity	150,000
Hub Equipment	383,250	Venture Capital	4,557,850
Admin Hardware	75,000	Net Profit	1,136,460
Total Non-Current Assets	508,250	Total Owner's Equity	5,794,310
Total Assets	5,794,310	Total Liabilities & Equity	5,794,310

Financial Model: Revenue & Arbitrage Engine

Commodity Arbitrage: Weighted avg market rate = 9,000 EGP/ton

BUY PRICE

6,300 EGP/ton

70% of market rate
Material Acquisition (COGS)

SELL PRICE

9,900 EGP/ton

110% of market rate
Gross Revenue per ton

GROSS MARGIN

3,600 EGP/ton

40 percentage-point spread
Structural margin on every ton

130 T/month

5T/day × 26 days

1,287,000 EGP

Monthly Gross Revenue

15,444,000 EGP

Annual Gross Revenue

1,136,460 EGP

Annual Net Profit (Year 1)

4,707,850 EGP

Total Start-up Capital

5,286,060 EGP

Cash at Year End

03

System Analysis

UML Diagrams: Use Case, Activity & Sequence

Use Case Diagrams — Five Actor Model

Supplier

- Register facility & sign contract
- View pickup schedules
- Track payout history
- ESG compliance page
- Chat with AI chatbot

Factory

- Browse baled commodity inventory
- Confirm purchase orders
- Schedule FTL delivery dates
- Track shipment status
- Access formal invoices

Driver

- View daily pickup routes
- Confirm arrivals at suppliers
- Log load details & photos
- Confirm hub arrival

Hub Manager

- Manage full logistics pipeline
- Two-stage weighing (QA)
- Update Algorithmic Pricing
- Oversee inventory & baling
- Daily report to Super Admin

Super Admin

- Full system-wide oversight
- Approve onboarding requests
- Manage users & roles
- Monitor audit logs
- Configure global parameters

Activity Diagrams — 4-Stage Operational Pipeline

Part 1: Registration & Contract Flow

Supplier/Factory submits registration → Super Admin reviews → schedules meeting → registers contract → creates pickup schedule → notifies user.

Part 2: Inbound Logistics (Driver)

Driver views routes → confirms arrival at supplier → loads waste & uploads photo/weight → departs toward hub → confirms hub arrival (notifies Hub Manager).

Part 3: Hub Processing (Stages 2 & 3)

Hub Manager: Tier 1 weigh → QA sorters decontaminate → Tier 2 weigh → contamination ratio check → price at 70% → bale → assign Bale ID → daily report.

Part 4: Outbound & Factory Confirmation (Stage 4)

24-ton threshold alert → Super Admin creates delivery note → Factory confirms receipt → 48-hr rejection window → Sales invoice at 110% → payment outside system.

Part 5: Common Activities (All Users)

All authenticated users can: Sign In / Register, Chat with AI chatbot, View profile details.

04

Database Design

Class Diagram, ERD & Database Mapping

System Design: Class Diagram & ERD

Class Diagram Entities

- User (base — all roles inherit)
- Supplier (CorporateGenerator)
- Factory (RecyclingBuyer)
- Driver / Loader
- HubManager
- SuperAdmin
- Contract
- PickupSchedule
- WeightReceipt
- BaleInventory
- PricingEngine
- SalesInvoice

ERD Key Relationships

User ————— has one Role

Supplier ————— has many Contracts

Contract ————— has many PickupSchedules

Driver ————— assigned many Routes

WeightReceipt — generated per Shipment

PricingEngine — computes per Commodity

BaleInventory — tracks per Commodity type

SalesInvoice — generated per FTL delivery

Factory ————— confirms many Deliveries

System Design: Class Diagram & ERD

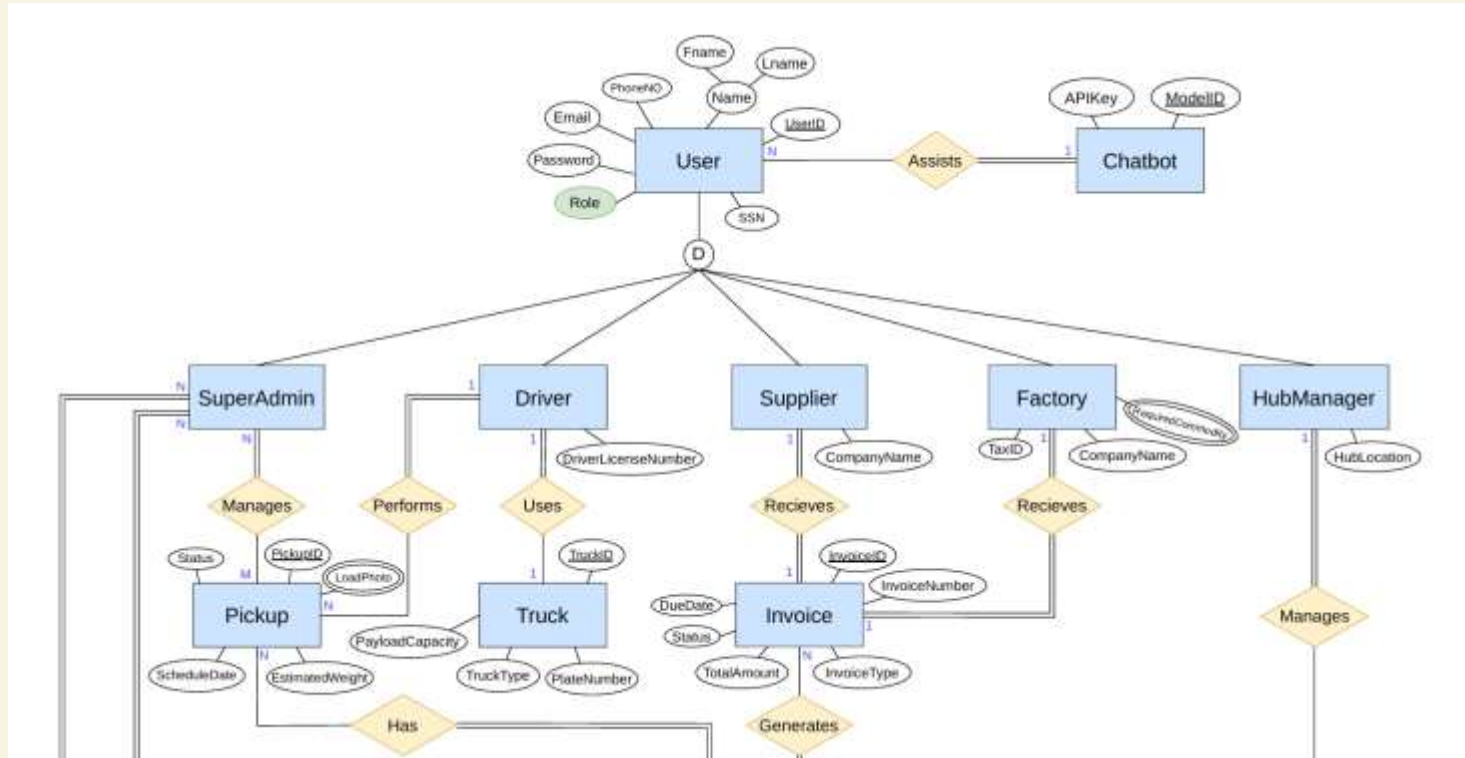
UserRole
SUPER_ADMIN
HUB_MANAGER
DRIVER
SUPPLIER
FACTORY

InvoiceStatus
DRAFT
PENDING
PAID
OVERDUE
DISPUTED

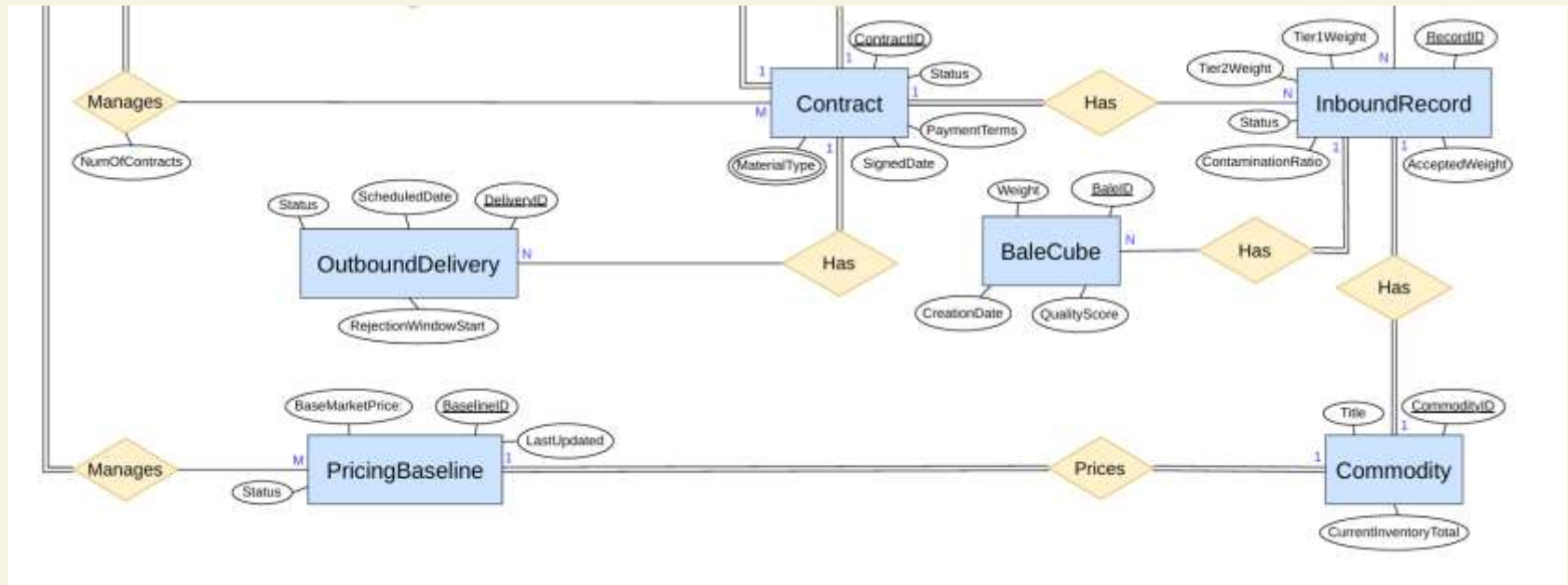
LoadStatus
RECEIVED
WEIGHTED_TIER1
SORTED
WEIGHTED_TIER2
BALED
READY_FOR_SHIPMENT

Commodity
PET
HDPE
PAPER
METAL
GLASS
MIXED_PLASTIC
OTHER

System Design: Class Diagram & ERD



System Design: Class Diagram & ERD



Database Architecture & Key Design Decisions

Role-Based Access Control (RBAC)

Single User table with role enum field. Each role maps to a distinct dashboard and permission set: Supplier, Factory, Driver, HubManager, SuperAdmin. All sensitive operations gate-checked server-side.

Algorithmic Pricing Engine

Admins set BaseMarketPrice per Commodity. Engine auto-computes $\text{BuyPrice} = 0.70 \times \text{base}$, $\text{SellPrice} = 1.10 \times \text{base}$. Price updates propagate to all dashboards in real-time without manual intervention.

Contamination Ratio Enforcement

Formula: $\text{ratio} = (\text{Tier1Weight} - \text{Tier2Weight}) / \text{Tier1Weight}$. If $\text{ratio} > 0.05$ (5%), AcceptedWeight is penalized; if $\text{ratio} > \text{threshold}$, full rejection is triggered and logged digitally.

Bale Traceability

Each bale is assigned a unique BaleID, commodity type, quality score, weight, and creation timestamp. Inventory tracked until 24-ton FTL threshold triggers outbound delivery workflow.

Outbound 48-Hour Window

On Factory delivery confirmation, system starts a 48-hour rejection clock. No rejection = auto-generate SalesInvoice at 110% rate. Admin manually marks invoice as paid after bank transfer.

05

System Implementation

Platform Architecture, Dashboards & Source Code

System Architecture Map

Supplier Portal

- Registration & contract signing
- View pickup schedules
- ESG compliance page
- Weight receipts & payout history
- AI chatbot (Gemini)

Hub Admin Console

- Inbound fleet scheduling
- QA weighbridge module
- Algorithmic Pricing Engine
- Inventory management
- Daily reports to Super Admin

Factory Portal

- Baled commodity inventory browse
- Confirm purchase orders
- Schedule FTL deliveries
- Sales invoice access
- Shipment tracking

SHARED BACKEND | React Frontend · PHP / Laravel API · SQL Server · Gemini AI Chatbot · AWS Cloud

06

Testing Methodology

Test Cases, Validation & Quality Assurance

Testing Strategy & Key Test Modules

Black-box & functional testing across all user roles; cross-browser (Chrome, Edge, Safari); mobile responsiveness (3G/4G); RBAC validation

Algorithmic Pricing Engine

Verify buy/sell rates auto-computed at 70%/110% of admin-set baselines. Ensure price changes propagate across all dashboards immediately with no manual intervention.

Outbound Logistics & Factory

Test 24-ton threshold alert, outbound delivery note creation, 48-hr rejection window enforcement, and Sales Invoice generation at 110% market rate.

Login & Sign Up Validation

Input validation: required fields, email format, password strength, duplicate account detection. Session management and secure logout tested.

Hub QA & Weighbridge

Validate Tier 1 / Tier 2 weighing sequence. Confirm contamination ratio formula triggers deduction or rejection at > 5% threshold. Verify digital weight receipts are generated.

Role-Based Access Control

Each user role must see only permitted dashboard sections. Attempt cross-role access with different session tokens and confirm all unauthorized access is blocked.

Core System Features

End-to-end flow tested: registration → contract → pickup schedule → inbound logistics → QA processing → inventory → outbound dispatch → invoice cycle.

07

Conclusion & Future Work

Impact, SDG Alignment & Roadmap

Project Achievements & Impact



Digital B2B Waste Platform Built

Three role-based web dashboards (Supplier, Hub Admin, Factory) fully developed with React + PHP/Laravel backend.



Algorithmic Pricing Engine Deployed

Zero-bidding system computing buy (70%) and sell (110%) rates in real-time. Structural 40pt margin locked on every ton.



Full QA Pipeline Implemented

Two-stage weighbridge, 5% contamination threshold, digital weight receipts, and Bale ID inventory tracking.



ESG Compliance Framework

MoE MoU-aligned ESG reporting page. Supports Egypt's Law 202/2020, Vision 2030, and UN SDG 12.



AI Chatbot Integration

Gemini-powered chatbot restricted to onboarding guidance, FAQ resolution, and waste classification support.



Viable Financial Model Validated

Year 1 net profit: 1,466,400 EGP. Cash at year end: 5,286,060 EGP. Break-even achievable within launch year.

Future Work Roadmap

Phase 2

Mobile App for Suppliers & Drivers

Native iOS/Android apps for real-time route tracking and instant weight receipt delivery.

Phase 4

Digital Payment Integration

Automate financial transactions between stakeholders, eliminating manual bank transfer tracking.

Phase 6

Blockchain Traceability

Immutable records for waste collection, processing, and delivery for enhanced ESG auditability.

Phase 8

E-Waste & Organic Waste Streams

Extend platform to cover electronic waste, organic waste, and industrial by-products.

Phase 3

Multi-Hub Expansion

Replicate hub model in Alexandria, Mansoura, and Upper Egypt. Centralized multi-hub management.

Phase 5

AI-Powered Predictive Analytics

ML models to forecast waste volumes, commodity demand, and pricing trends for proactive operations.

Phase 7

Automated ESG Reporting

Auto-generate dynamic sustainability reports using live platform data for corporate clients.

Phase 9

AI Decision Support System

Advanced AI for inventory forecasting, operations optimization, and profitability improvement.

Thank You

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Questions & Discussion

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